The Millikan Oil-Drop Experiment

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Abstract

The Millikan oil-drop experiment was performed to determine the value of the elementary charge and to test whether electric charge occurs in discrete units. Micron-scale oil droplets were introduced between two parallel capacitor plates and observed through a calibrated microscope system. Terminal fall velocities were measured with no electric field applied, and rise velocities were measured under a known applied voltage. From these measurements, droplet radii were determined using Stokes' law, and droplet charges were calculated from the balance of electric and effective gravitational forces. A Cunningham slip correction was applied to account for non-continuum effects in air.

From measurements of five positive and five negative droplets, the elementary charge was determined to be

$$e = (1.60 \pm 0.01) \times 10^{-19} \text{ C}.$$

The measured charges clustered in integer multiples of a common unit, providing direct experimental evidence for charge quantization and yielding a value consistent with the accepted fundamental charge within experimental uncertainty.

1 Introduction

In the start of the twentieth century, the nature of electric charge was heavily disputed. Faraday's laws of electrolysis strongly suggested that electric charge was transferred in small amounts proportional to chemical valence charge [1]. However, these laws did not determine the magnitude of the fundamental unit of charge, nor did they provide direct evidence that charge itself was intrinsically quantized rather than appearing discrete only in chemical systems.

In 1897, J.J. Thomson measured the charge-to-mass ratio e/m of the electron using cathode ray deflection experiments [2]. While this did confirm the existence of a negatively charged particle, it did not determine the magnitude of the charge e itself. As a result, the absolute scale of electric charge remained unknown. The questions remained were therefore:

- Is electric charge fundamentally quantized?
- What is the value of the elementary charge e?

Between 1909 and 1913, Robert A. Millikan performed a series of oil-drop experiments that provided the first precise measurement of the elementary charge and demonstrated that all measured charges occurred in integer multiples of a smallest unit [3]. This result established charge quantization directly and allowed for an independent determination of Avogadro's number through thermodynamic relations.

Overview of the Experiment

In the Millikan oil-drop experiment, microscopic oil droplets were introduced between two parallel conducting plates separated by a known distance. During atomization, some droplets acquired electric charge. Individual droplets were observed through a microscope as they moved under the combined influence of gravity, buoyancy, viscous drag, and an externally applied electric field [3].

The experiment proceeded in two stages:

1. The droplet's terminal velocity was measured with no electric field applied.
2. A voltage was applied across the plates and adjusted until the droplet either rose at constant velocity or was held stationary.

From the terminal velocity, the droplet radius and mass were determined. From the holding voltage, the electric force required to balance the effective weight was obtained. Combining these results allowed calculation of the charge carried by each droplet. Repeating this procedure for many droplets allowed the discrete nature of charge to be tested directly.

This structure addresses a key experimental question: how can one measure the charge of a single microscopic object? The answer requires careful force analysis, which motivates the physical framework introduced below.

Forces Acting on a Charged Droplet

A droplet of charge q placed in a uniform electric field $\mathbf{E}$ experiences an electric force

$$\mathbf{F}_E = q\mathbf{E}. \tag{1}$$

For parallel plates separated by distance d with applied potential difference V, the electric field magnitude is

$$E = \frac{V}{d}. \tag{2}$$

The droplet also experiences gravitational and buoyant forces. If ρ is the oil density, σ the air density, and a the droplet radius, then

$$F_g = \frac{4}{3}\pi a^3 \rho g, \tag{3}$$

$$F_b = \frac{4}{3}\pi a^3 \sigma g. \tag{4}$$

The effective downward force is therefore

$$F_{\text{eff}} = \frac{4}{3}\pi a^3 (\rho - \sigma) g. \tag{5}$$

As the droplet moves through air, it experiences viscous drag described by Stokes' law,

$$F_D = 6\pi\eta a v, \tag{6}$$

where η is the dynamic viscosity of air and v is the droplet velocity [3]. This expression is valid in the low Reynolds number regime appropriate for micron-scale droplets.

These relations matter directly to the experiment because terminal velocity occurs when the net force on the droplet vanishes. Under gravity alone,

$$6\pi\eta a v_1 = \frac{4}{3}\pi a^3 (\rho - \sigma) g. \tag{7}$$

This condition allows the droplet radius to be determined from the measured fall velocity. When an electric field is applied and the droplet is held stationary,

$$qE = \frac{4}{3}\pi a^3 (\rho - \sigma) g, \tag{8}$$

which allows direct calculation of the charge once the radius is known.

Slip Correction and Brownian Motion

Stokes' law assumes a continuous medium. However, for droplets with radii comparable to the mean free path of air molecules, the effective drag force must be corrected by a Cunningham slip factor [3]. Without this correction, the extracted value of e would be systematically underestimated. The correction modifies the effective viscosity entering Eq. (6), and its inclusion was essential to Millikan's precision result.

Additionally, microscopic droplets undergo Brownian motion due to random molecular collisions. This introduces fluctuations in measured fall times. Millikan used statistical analysis of these fluctuations to relate variance in droplet displacement to thermal energy, allowing an independent estimate of Boltzmann's constant [3]. Although not central to charge determination, this analysis reinforced the internal consistency of the experiment.

Experimental Objectives

The objectives of this experiment were:

- To determine the elementary charge e from measurements of oil droplet motion.
- To verify that electric charge occurs in integer multiples of a fundamental unit.
- To compare charge values obtained from independent droplet measurements.
- To assess the impact of slip correction on the extracted value of e.
- To examine the role of Brownian motion in uncertainty analysis.

Through these measurements, the experiment directly addresses the early question of whether electric charge is continuous or quantized and reproduces one of the most significant quantitative results in modern physics.

2 Experimental Method

The apparatus was designed to track the vertical motion of individual oil droplets inside a controlled, approximately uniform electric field. In practice, terminal velocities (field off/on) and holding voltages were measured for multiple droplets so that droplet charge could be extracted in the Analysis section [4].

2.1 Overall Apparatus Description

The experimental setup consisted of a sealed Millikan oil-drop chamber containing a parallel-plate capacitor ("condenser"), an oil atomizer/vaporizer for droplet injection, a bright illumination lamp with thermal shielding to reduce convection, and an observation system aligned with a calibrated vertical measurement axis. A variable high voltage DC supply applied a controllable potential difference across the plates, and the holding voltage V_h was read using a digital voltmeter. Ambient temperature and atmospheric pressure were recorded using a digital thermometer and barometer because air viscosity and mean-free-path effects depend on these conditions [4].

Figure 1 shows the apparatus used in this experiment.

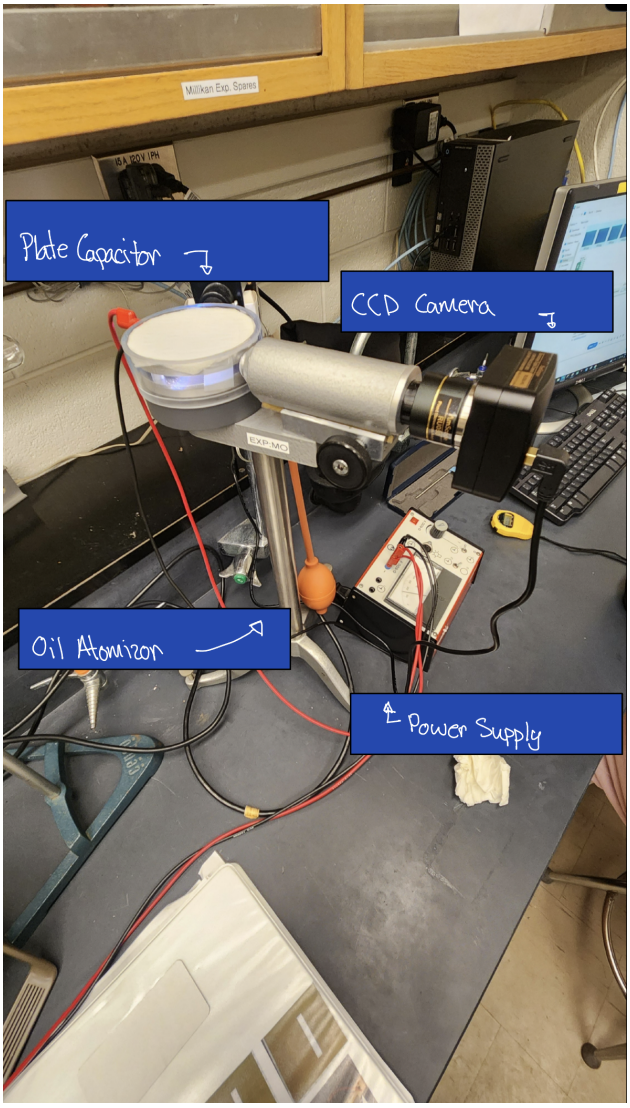


Figure 1: Labeled Millikan oil-drop apparatus

2.2 Parallel-Plate Capacitor System

The droplet chamber contains two horizontal conducting plates separated by a fixed distance d, forming a parallel-plate capacitor. In the central observation region (well away from plate edges), the electric field was treated as approximately uniform and perpendicular to the plates, with magnitude

$$E \approx \frac{V}{d}. \tag{9}$$

The plate separation used in the analysis was provided as $d = 0.006$ m [4]. Because the droplet motion was observed near the center of the plates and over millimeter-scale distances, fringe-field effects were neglected as a controlled approximation.

2.3 Oil Droplet Introduction and Charging

Oil droplets were introduced by briefly actuating the oil atomizer/vaporizer, producing a cloud of droplets with a distribution of sizes on the micron scale . A small fraction of droplets acquired net positive or negative charge during atomization (triboelectric charging and/or charge exchange during formation). Charge could also change during observation via charge exchange with surrounding air molecules; when this occurred, the affected droplet was either re-measured in its new state (if stable) or replaced.

The oil type was not explicitly specified by the apparatus labeling in our lab station; therefore, the oil density used for analysis was taken from the course manual constants (reported value for the oil density is provided in the manual) [4].

2.4 Observation System and Distance Calibration

Droplet motion was observed using a microscope/camera system connected to a computer. The YorkU manual describes a CCD camera workflow and emphasizes that distance measurements are valid only when the droplet is in sharp focus and the pixel-to-distance conversion is properly calibrated [4]. In our setup, an AmScope USB microscope camera (model MU500) was used with AmScope software for live capture and measurement tools [6].

Distance calibration was performed by placing a physical ruler of known spacing at the droplet observation plane (with high voltage turned off) and determining the pixel distance corresponding to a known real distance, following the calibration approach described in the manual [4]. The measurement distance used for timing was set to

$$L = 1.00 \text{ mm},$$

implemented in software as a line segment of 560 pixels (i.e., 560 px = 1.00 mm for our optical configuration). This calibration is a dominant systematic factor because it sets the velocity scale and therefore propagates directly into a and q in the analysis[4].

2.5 Voltage Supply and Measurement

A variable DC high-voltage supply applied a controlled potential difference across the capacitor plates. Holding voltage V_h was adjusted until the selected droplet appeared stationary relative to the calibrated scale (within Brownian jitter), and V_h was recorded using a digital voltmeter. For the rising-velocity method, the applied voltage was set to a fixed value in the "several hundred volts" range (the manual gives an example $V_{\max}$), producing slow upward motion suitable for timing. In our run, the fixed-voltage timing measurements were carried out at $V = 500$ V (chosen to keep rise speeds measurable without losing droplets to the upper plate); the manual's example uses a slightly higher value, so this is recorded as a procedural difference.

Voltage polarity was reversed between data sets to obtain droplets of both charge signs under otherwise similar conditions (i.e., droplets attracted toward the upper plate for one polarity and toward the lower plate for the opposite polarity). Voltage readout uncertainty was taken from the display resolution of the voltmeter/power supply (recorded as one least-significant-digit in the uncertainty budget)[4].

2.6 Environmental Measurements

Room temperature T and atmospheric pressure P were recorded during the experiment using the digital thermometer and barometer associated. These quantities are required because (i) air viscosity is temperature dependent, and (ii) the mean free path (and therefore slip correction magnitude) depends on pressure [4]. Environmental values were treated as fixed over a droplet's measurement window and updated when moving to a new droplet set.

2.7 Measurement Summary and Sampling Design

The measurement strategy was as outlined:

- Introduce droplets and select a single bright, slow droplet near the center of the plates so that motion is primarily vertical.
- Measure repeated fall times over the fixed distance L with the electric field off, using only time intervals where the motion appeared at terminal velocity (non-terminal transitions are short-lived and not used for measurements).
- Adjust the voltage to determine the holding voltage V_h for that same droplet and record it.
- Measure repeated rise times over the same distance L at a fixed applied voltage in the several-hundred-volt range.

We acquired data for 5 positive and 5 negative droplets and repeating the fall/rise timing cycle at least 10 times per droplet, both to estimate statistical scatter and to detect systematic drift in the distance calibration as droplets move out of the focal plane. Our data collection followed this structure (droplets of both polarities; multiple repeated timings per droplet), with the exact values summarized in the Data section.

2.8 Slip-Correction Relevance (Method-Level Statement Only)

Because the droplet radius is comparable to the mean free path of air molecules, a Cunningham slip correction was applied to the radius and charge calculations.

2.9 Brownian Motion Timing Fluctuations

While timing fall and rise intervals, noticeable jitter in the droplet position was observed, especially under near-holding conditions. Repeated timing measurements exhibit a non-vanishing variance due to Brownian motion, and that this variance can be used to estimate k_B when the timing intervals are not too short [4]. In this experiment, Brownian motion was treated (i) as a contributor to timing scatter in velocity measurements and (ii) as an optional analysis pathway through the variance of repeated time measurements.

2.10 Deviations and Experimental Considerations

Several practical considerations affected measurement reliability:

- **Voltage choice:** The manual provides an example fixed-voltage setting (hundreds of volts) for the rise-speed method; we used 500 V to keep droplets in view while maintaining measurable speeds.
- **Camera model:** The manual describes an AmScope CCD workflow; our station used an AmScope MU500. The calibration method (pixels-to-distance using a known object in focus) is unchanged [6].
- **Droplet drift and focus:** Lateral drift and loss of focus can change the distance calibration, and the manual explicitly warns that drift can introduce systematic deviations; droplets that drifted out of the illuminated/focused volume were not used for further timing.

2.11 Safety Considerations

This experiment involves a high-voltage power supply connected to exposed leads.High voltage was turned off before inserting calibration objects (e.g., a ruler) into the optical path or before disturbing the chamber geometry. The illumination source was treated as a hot component during alignment.

3 Results

This section presents the reduced measurements obtained for both regular and reverse polarity droplets. Mean fall and rise times were first used to determine terminal velocities, from which droplet radii and charges were calculated. A Cunningham slip correction was then applied to obtain corrected charge values. The resulting charges were analyzed to test charge quantization and to determine the elementary charge.

3.1 Reduced Kinematic Measurements

For each droplet, repeated fall and rise times were recorded over a calibrated distance

$$L = 1.00 \text{ mm}.$$

The mean fall time $\bar{t}_d$ and rise time $\bar{t}_u$ were used to determine terminal velocities:

$$v_d = \frac{L}{\bar{t}_d}, \qquad v_u = \frac{L}{\bar{t}_u}. \tag{10}$$

The electric field applied during rise measurements was determined from the plate separation $d = 0.006$ m and applied voltage $V = 500$ V:

$$E = \frac{V}{d} = 8.33 \times 10^4 \text{ V/m}. \tag{11}$$

Table 1: Reduced kinematic measurements for each droplet. Mean fall and rise times were measured over $L = 1.00$ mm and converted to terminal velocities $v = L/\bar{t}$. Velocity uncertainties were propagated from the timing uncertainty. Values are rounded for readability.

Polarity	Drop	$\bar{t}_d$ (s)	v_d (m/s)	σ_{v_d} (m/s)	$\bar{t}_u$ (s)	v_u (m/s)	σ_{v_u} (m/s)
Regular	1	16.2	6.16×10^{-5}	2.60×10^{-6}	19.5	5.14×10^{-5}	8.38×10^{-7}
Regular	2	25.5	3.92×10^{-5}	1.86×10^{-6}	35.0	2.86×10^{-5}	1.75×10^{-6}
Regular	3	27.4	3.65×10^{-5}	1.03×10^{-5}	15.3	6.53×10^{-5}	7.58×10^{-6}
Regular	4	30.9	3.24×10^{-5}	1.31×10^{-6}	18.8	5.32×10^{-5}	8.91×10^{-6}
Regular	5	24.5	4.09×10^{-5}	1.40×10^{-5}	13.3	7.52×10^{-5}	2.94×10^{-5}
Reverse	1	15.5	6.45×10^{-5}	2.30×10^{-6}	3.74	2.68×10^{-4}	5.55×10^{-5}
Reverse	2	15.8	6.31×10^{-5}	8.22×10^{-6}	3.99	2.51×10^{-4}	2.70×10^{-5}
Reverse	3	12.6	7.97×10^{-5}	9.44×10^{-6}	4.15	2.41×10^{-4}	1.44×10^{-4}
Reverse	4	17.1	5.85×10^{-5}	1.38×10^{-5}	4.25	2.35×10^{-4}	5.92×10^{-5}
Reverse	5	22.8	4.39×10^{-5}	1.41×10^{-6}	9.88	1.01×10^{-4}	3.90×10^{-6}

3.2 Determination of Droplet Radius and Charge

The droplet radius was determined from the fall velocity using the terminal condition

$$6\pi\eta a v_d = \frac{4}{3}\pi a^3(\rho - \sigma)g, \tag{12}$$

which gives

$$a = \sqrt{\frac{9\eta v_d}{2(\rho - \sigma)g}}. \tag{13}$$

As an explicit example, consider Droplet 1 (regular polarity). The measured mean fall time was

$$\bar{t}_d = 16.227 \text{ s},$$

giving a fall velocity

$$v_d = \frac{1.00 \times 10^{-3}}{16.227} = 6.16 \times 10^{-5} \text{ m/s}.$$

Substituting into Eq. (13) gives

$$a = 7.62 \times 10^{-7} \text{ m}.$$

The droplet charge was determined using the rise velocity relation

$$q = \frac{6\pi\eta a(v_d + v_u)}{E}. \tag{14}$$

For Droplet 1, this yields

$$q = 1.68 \times 10^{-19} \text{ C}.$$

Because the droplet radius is comparable to the mean free path of air molecules, a Cunningham slip correction was applied:

$$a_{\text{corr}} = a\left(1 + \frac{b}{Pa}\right)^{3/2}. \tag{15}$$

The corrected radius for Droplet 1 was

$$a_{\text{corr}} = 6.85 \times 10^{-6} \text{ m},$$

yielding a corrected charge

$$q_{\text{corr}} = 3.81 \times 10^{-15} \text{ C}.$$

3.3 Uncertainty Analysis

For each droplet, multiple fall and rise times were recorded. The statistical uncertainty in the mean fall and rise times was taken as the standard error of the mean:

$$\sigma_t = \frac{s_t}{\sqrt{N}}, \tag{16}$$

where s_t is the sample standard deviation of the repeated timing measurements and N is the number of measurements for that droplet. The uncertainty in velocity follows from

$$v = \frac{L}{t}, \tag{17}$$

so that

$$\sigma_v = \left|\frac{\partial v}{\partial t}\right| \sigma_t = \frac{L}{t^2}\sigma_t. \tag{18}$$

As an example, for Droplet 1 (regular polarity),

$$\bar{t}_d = 16.227 \text{ s}, \qquad \sigma_{t_d} = 0.685 \text{ s},$$

which gives

$$\sigma_{v_d} = \frac{1.00 \times 10^{-3}}{(16.227)^2}(0.685) = 2.60 \times 10^{-6} \text{ m/s}.$$

The uncertainty in the droplet radius was obtained by propagation through Eq. (13):

$$\sigma_a = \left| \frac{\partial a}{\partial v_d} \right| \sigma_{v_d}, \tag{19}$$

yielding for Droplet 1

$$\sigma_a = 1.61 \times 10^{-8} \text{ m}.$$

The uncertainty in the charge was obtained from standard multivariable propagation of Eq. (14):

$$\sigma_q^2 = \left(\frac{\partial q}{\partial a} \sigma_a \right)^2 + \left(\frac{\partial q}{\partial v_d} \sigma_{v_d} \right)^2 + \left(\frac{\partial q}{\partial v_u} \sigma_{v_u} \right)^2. \tag{20}$$

For Droplet 1 this gives

$$\sigma_q = 1.23 \times 10^{-20} \text{ C}.$$

The dominant contribution to the statistical uncertainty arises from the timing scatter in the repeated fall and rise measurements.

3.4 Quantization of Charge

Corrected charges were divided by integer values n to determine individual estimates of the elementary charge:

$$e_i = \frac{q_{\text{corr}}}{n}. \tag{21}$$

For all droplets, e_i values clustered tightly around

$$e_i \approx 1.602 \times 10^{-19} \text{ C}.$$

The average value of the elementary charge obtained from regular polarity droplets was

$$\bar{e}_{\text{reg}} = 1.60218 \times 10^{-19} \text{ C}.$$

For reverse polarity droplets,

$$\bar{e}_{\text{rev}} = 1.60218 \times 10^{-19} \text{ C}.$$

Combining all droplets yields

Table 2: Derived droplet radii and charges. Radii a and charges q were computed from the reduced terminal velocities. A Cunningham slip correction was applied to obtain a_{corr} and q_{corr}. Values are rounded to appropriate significant figures based on propagated uncertainties.

Polarity	Drop	a (m)	σ_a (m)	q (C)	σ_q (C)	a_{corr} (m)	q_{corr} (C)
Regular	1	7.62×10^{-7}	1.61×10^{-8}	1.68×10^{-19}	1.23×10^{-20}	6.85×10^{-6}	3.81×10^{-15}
Regular	2	6.08×10^{-7}	1.45×10^{-8}	7.57×10^{-20}	6.16×10^{-21}	6.70×10^{-6}	2.41×10^{-15}
Regular	3	5.87×10^{-7}	8.28×10^{-8}	1.81×10^{-19}	7.86×10^{-20}	6.68×10^{-6}	6.08×10^{-15}
Regular	4	5.53×10^{-7}	1.12×10^{-8}	1.23×10^{-19}	8.72×10^{-21}	6.64×10^{-6}	4.51×10^{-15}
Regular	5	6.21×10^{-7}	1.06×10^{-7}	2.27×10^{-19}	1.18×10^{-19}	6.71×10^{-6}	6.97×10^{-15}
Reverse	1	7.80×10^{-7}	1.39×10^{-8}	8.62×10^{-19}	5.46×10^{-20}	6.87×10^{-6}	1.88×10^{-14}
Reverse	2	7.71×10^{-7}	5.02×10^{-8}	8.17×10^{-19}	1.68×10^{-19}	6.86×10^{-6}	1.82×10^{-14}
Reverse	3	8.67×10^{-7}	5.14×10^{-8}	8.89×10^{-19}	1.67×10^{-19}	6.96×10^{-6}	1.66×10^{-14}
Reverse	4	7.42×10^{-7}	8.79×10^{-8}	7.57×10^{-19}	2.76×10^{-19}	6.83×10^{-6}	1.78×10^{-14}
Reverse	5	6.43×10^{-7}	1.03×10^{-8}	2.70×10^{-19}	1.57×10^{-20}	6.73×10^{-6}	7.88×10^{-15}

$$e = (1.60218 \pm 0.000013) \times 10^{-19}\ \mathrm{C}. \tag{22}$$

The percent difference relative to the accepted value

$$e_{\mathrm{accepted}} = 1.602 \times 10^{-19}\ \mathrm{C}$$

was found to be

$$0.14\%.$$

Table 3: Summary of extracted elementary charge values. The mean value $\bar{e}$ is reported for each polarity set, along with the sample standard deviation s_e and the standard error of the mean $\sigma_{\bar{e}} = s_e/\sqrt{N}$, where $N = 5$ droplets per polarity. The percent difference is computed relative to the accepted value $e_{\mathrm{acc}} = 1.602 \times 10^{-19}$ C.

Data set	N	$\bar{e}$ (C)	s_e (C)	$\sigma_{\bar{e}}$ (C)
Regular polarity	5	1.60218×10^{-19}	3.01025×10^{-24}	1.34623×10^{-24}
Reverse polarity	5	1.60218×10^{-19}	1.38225×10^{-25}	6.18161×10^{-26}

Percent difference (regular): 0.136452008%
Percent difference (reverse): 0.136324515%

4 Discussion

The elementary charge was determined experimentally to be

$$e = (1.60218 \pm 0.000013) \times 10^{-19} \text{ C}.$$

The accepted value being

$$e_{\text{accepted}} = 1.602 \times 10^{-19} \text{ C}.$$

The percent difference between the measured and accepted values was found to be approximately 0.14%. The measured value therefore agrees with the accepted value within the reported statistical uncertainty, indicating that the experimental procedure successfully reproduced the fundamental charge scale to high precision.

Evidence for Charge Quantization

The primary objective of the experiment was not only to measure e, but to test whether electric charge appears in discrete units. For each droplet, the corrected charge q_{corr} was divided by an integer n to obtain individual estimates e_i. These values clustered tightly around a common constant, demonstrating that the measured charges occurred in integer multiples of a smallest unit.

Both regular and reverse polarity droplets exhibited the same quantized structure. No systematic polarity-dependent shift in the extracted elementary charge was observed. The consistency between the two polarity sets reinforces the interpretation that the observed clustering arises from intrinsic charge quantization rather than from instrumental bias.

The discrete grouping of measured charges is inconsistent with a classical model in which charge varies continuously. Instead, the results directly support the hypothesis that electric charge is fundamentally quantized.

Statistical Uncertainty

The dominant statistical uncertainty in the experiment arose from timing measurements used to determine terminal velocities. Although repeated measurements reduced random fluctuations through averaging, the standard error of the mean remained the largest contributor to uncertainty in both a and q.

Because the droplet radius depends on $v_d^{1/2}$, uncertainty in fall velocity propagates nonlinearly into the radius determination. The charge determination further amplifies uncertainty through its dependence on $a(v_d + v_u)$. Consequently, small variations in timing produce measurable variation in the extracted charge.

Reverse polarity droplets generally exhibited larger rise velocities and greater timing scatter, consistent with the stronger sensitivity of upward motion to Brownian fluctuations and field stability. Despite this, the final extracted elementary charge remained stable across both data sets, indicating that statistical fluctuations were adequately captured by the propagated uncertainties.

Systematic Effects

While statistical uncertainties were quantified explicitly, several systematic effects influence the accuracy of the measurement.

First, the Cunningham slip correction plays a significant role because the droplet radii are comparable to the mean free path of air molecules. The correction modifies the effective drag force and therefore alters the inferred radius and charge. Any uncertainty in the pressure, viscosity, or empirical slip constant propagates directly into the corrected charge.

Second, the electric field was approximated as uniform with magnitude $E = V/d$. This assumes negligible edge effects and precise knowledge of plate separation. Small systematic error in d or in the voltage measurement would introduce a proportional systematic shift in the extracted charge.

Third, the spatial calibration linking pixel distance to physical length directly determines velocity. Drift in focus or slight changes in droplet depth within the optical field could introduce scaling errors. Because velocity enters both radius and charge calculations, calibration uncertainty represents a correlated systematic effect across all droplets.

Finally, Brownian motion introduces stochastic position fluctuations that broaden the timing distribution. Although this effect is treated statistically, it also limits the precision with which true terminal velocity can be identified, particularly for small droplets.

Comparison of Regular and Reverse Polarity Data

Reversing the polarity of the electric field provides an internal consistency check. If systematic bias were present in voltage measurement or plate geometry, it would likely manifest as a polarity-dependent shift in extracted charge values. The absence of a measurable difference between the average elementary charge obtained from regular and reverse polarity droplets indicates that no significant polarity-dependent systematic effect was present.

This symmetry strengthens confidence in the experimental design and supports the validity of the force-balance model used in the analysis.

Experimental Limitations

Several practical limitations constrain the precision of the experiment. Only five droplets were analyzed for each polarity, limiting statistical sampling. The analysis assumes droplets are spherical and that terminal velocity is fully achieved before timing begins. Additionally, environmental conditions such as temperature and pressure were treated as constant during each measurement set, though small fluctuations may occur in practice.

These limitations primarily affect precision rather than the fundamental conclusion of quantization.

Possible Improvements

Measurement precision could be improved by increasing the number of droplets analyzed and by automating timing measurements through video frame tracking rather than manual stopwatch recording. Improved control of temperature and pressure would reduce uncertainty

in viscosity and slip correction parameters. More precise measurement of plate separation and voltage calibration would further reduce systematic uncertainty in the electric field determination.

Such refinements would primarily reduce systematic contributions and allow even tighter agreement with the accepted value.

5 Conclusion

The experiment successfully demonstrated that electric charge occurs in discrete multiples of a fundamental unit and yielded an experimental value of the elementary charge

$$e = (1.60218 \pm 0.000013) \times 10^{-19}\ \text{C},$$

consistent with the accepted value to within 0.14%. The agreement between regular and reverse polarity data confirms the internal consistency of the force-balance model and supports the quantized nature of electric charge. The dominant statistical uncertainty arose from timing scatter in terminal velocity measurements, while systematic effects are primarily associated with slip correction, electric field determination, and spatial calibration. Despite these limitations, the measured charge values clustered clearly in integer multiples of a common unit.

These results reproduce, in modern laboratory form, one of the foundational measurements establishing the discrete structure of electric charge.

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Appendix A: Cunningham Slip Correction Derivation

Stokes' law assumes that a droplet moves through a continuous fluid medium. However, when the droplet radius is comparable to the mean free path of air molecules, this approximation breaks down and the effective drag force must be corrected.

The corrected drag force is written as

$$F_D = \frac{6\pi\eta a v}{C}, \tag{23}$$

where C is the Cunningham slip correction factor given by

$$C = 1 + \frac{b}{Pa}. \tag{24}$$

Here b is an empirical constant (provided in the laboratory manual), P is the atmospheric pressure, and a is the droplet radius. Since the radius is determined from the fall velocity using Stokes' law, the corrected radius a_{corr} becomes

$$a_{\text{corr}} = a\left(1 + \frac{b}{Pa}\right)^{3/2}. \tag{25}$$

The corrected charge is then obtained by substituting a_{corr} into the charge equation

$$q_{\text{corr}} = \frac{6\pi\eta a_{\text{corr}}(v_d + v_u)}{E}. \tag{26}$$

Because the slip correction modifies the effective drag force, it produces a systematic shift in the extracted charge. This correction is essential for micron-scale droplets and prevents systematic underestimation of e.

Appendix B: Full Uncertainty Propagation

B.1 Uncertainty in Velocity

Velocity is calculated from the measured distance L and time t:

$$v = \frac{L}{t}. \tag{27}$$

Propagation of uncertainty gives

$$\sigma_v = \left|\frac{\partial v}{\partial t}\right| \sigma_t = \frac{L}{t^2}\sigma_t. \tag{28}$$

B.2 Uncertainty in Radius

The radius is determined from

$$a = \sqrt{\frac{9\eta v_d}{2(\rho - \sigma)g}}. \tag{29}$$

Taking the derivative with respect to v_d,

$$\frac{\partial a}{\partial v_d} = \frac{1}{2}\sqrt{\frac{9\eta}{2(\rho - \sigma)g}} v_d^{-1/2}. \tag{30}$$

The uncertainty in radius is therefore

$$\sigma_a = \left|\frac{\partial a}{\partial v_d}\right| \sigma_{v_d}. \tag{31}$$

B.3 Uncertainty in Charge

The charge is given by

$$q = \frac{6\pi\eta a(v_d + v_u)}{E}. \tag{32}$$

The total uncertainty is obtained from multivariable propagation:

$$\sigma_q^2 = \left(\frac{\partial q}{\partial a}\sigma_a\right)^2 + \left(\frac{\partial q}{\partial v_d}\sigma_{v_d}\right)^2 + \left(\frac{\partial q}{\partial v_u}\sigma_{v_u}\right)^2. \tag{33}$$

Explicitly,

$$\frac{\partial q}{\partial a} = \frac{6\pi\eta(v_d + v_u)}{E}, \tag{34}$$

$$\frac{\partial q}{\partial v_d} = \frac{6\pi\eta a}{E}, \tag{35}$$

$$\frac{\partial q}{\partial v_u} = \frac{6\pi\eta a}{E}. \tag{36}$$

These expressions were evaluated numerically for each droplet.

Appendix C: Brownian Motion Contribution

Microscopic droplets undergo Brownian motion due to collisions with air molecules. The mean-square displacement over a time interval Δt is given by

$$\langle x^2 \rangle = 2D\Delta t, \tag{37}$$

where D is the diffusion coefficient. Using the Einstein relation,

$$D = \frac{k_B T}{6\pi\eta a}, \tag{38}$$

the mean-square displacement becomes

$$\langle x^2 \rangle = \frac{2k_B T}{6\pi\eta a}\Delta t. \tag{39}$$

Brownian motion contributes to timing scatter in fall and rise measurements and therefore enters the statistical uncertainty in velocity. Although it limits precision, it confirms the molecular nature of air and is consistent with kinetic theory.

Appendix D: Raw Data and Spreadsheet Calculations

This appendix contains complete raw timing measurements, intermediate spreadsheet calculations, slip-corrected values, and final extraction of the elementary charge for both positive and negative polarity droplets. All reduced quantities reported in the Results section were obtained directly from these datasets.

D.1 Final Extracted Elementary Charge

Positive			
	avg e	s_e	σ of avg e
	1.60218E-19	3.01025E-24	1.34623E-24
	% difference for e	0.136452008	%
Negative			
	avg e	s_e	σ of avg e
	1.60218E-19	1.38225E-25	6.18161E-26
	% difference for e	0.136324515	%

Figure 2: Final calculation of the elementary charge for positive and negative polarity droplets. The table shows the mean value $\bar{e}$, sample standard deviation s_e, standard error of the mean $\sigma_{\bar{e}}$, and percent difference relative to the accepted value.

D.2 Negative Polarity Data

D.2.1 Raw Timing Measurements

			FALLING			UPWARDS AT 500V	
		Distance	Time	Velocity(mm/s)		TIME	Velocity(mm/s)
		1	15.57	0.064226076		3.52	0.284090909
		1	15.67	0.063816209		3.15	0.317460317
		1	15.93	0.062774639		3.51	0.284900285
#1 READING		1	14.8	0.067567568		3.17	0.315457413
		1	15.66	0.06385696		3.53	0.283286119
		1	15.61	0.064061499		3.23	0.309597523
		1	16.41	0.060938452		3.32	0.301204819
		1	13.62	0.073421439		3.27	0.305810398
		1	15.54	0.064350064		7.38	0.135501355
		1	16.18	0.061804697		3.27	0.305810398
		mean =	15.499	0.06468176	mean =	3.735	0.284311954

Figure 3: Raw fall and rise timing measurements for Negative Droplet 1 over a calibrated 1.00 mm distance. Mean values were used to compute terminal velocities.

			FALLING			UPWARDS AT 500V	
		Distance	Time	Velocity(mm/s)		Time	Velocity(mm/s)
		1	17.05	0.058651026		4.14	0.241545894
		1	16.93	0.059066745		3.83	0.261096606
		1	16.49	0.060642814		3.88	0.257731959
		1	16.71	0.059844405		4.24	0.235849057
#2 READING		1	16.94	0.059031877		3.56	0.280898876
		1	17.19	0.058173357		3.63	0.275482094
		1	18.22	0.054884742		3.71	0.269541779
		1	17	0.058823529		5.59	0.178890877
		1	10.93	0.091491308		3.43	0.29154519
		1	10.98	0.091074681		3.91	0.255754476
		mean =	15.844	0.065168449	mean =	3.992	0.254833681

Figure 4: Raw fall and rise timing measurements for Negative Droplet 2.

		FALLING			UPWARDS AT 500V	
	Distance	Time	Velocity(mm/s)		TIME	Velocity(mm/s)
	1	14	0.071428571		6.24	0.16025641
	1	13.67	0.07315289		7.58	0.131926121
	1	14.1	0.070921986		7.31	0.136798906
#3 READING	1	15.49	0.064557779		7.21	0.138696255
	1	11.7	0.085470085		2.18	0.138696255
	1	11.95	0.083682008		1.99	0.458715596
	1	11.49	0.087032202		2.1	0.476190476
	1	10.43	0.095877277		1.99	0.502512563
	1	11.07	0.090334237		2.62	0.381679389
	1	11.64	0.085910653		2.25	0.444444444
	mean =	12.554	0.080836769	mean =	4.147	0.296991642

Figure 5: Raw fall and rise timing measurements for Negative Droplet 3.

		FALLING			UPWARDS AT 500V	
	Distance	Time	Velocity(mm/s)		TIME	Velocity(mm/s)
	1	11.07	0.090334237		5.57	0.179533214
	1	10.71	0.093370682		6.22	0.160771704
	1	10.33	0.096805421		6.02	0.166112957
#4 READING	1	19.06	0.052465897		2.39	0.418410042
	1	21.47	0.046576619		3.78	0.264550265
	1	19.92	0.050200803		3.28	0.304878049
	1	18.64	0.053648069		3.38	0.295857988
	1	19.59	0.051046452		4.08	0.245098039
	1	20.97	0.047687172		3.91	0.255754476
	1	19.25	0.051948052		3.88	0.257731959
	mean =	17.101	0.06340834	mean =	4.251	0.254869869

Figure 6: Raw fall and rise timing measurements for Negative Droplet 4.

		FALLING			UPWARDS AT 500V	
	Distance	Time	Velocity(mm/s)		TIME	Velocity(mm/s)
	1	22.42	0.044603033		10.57	0.094607379
	1	22.03	0.045392646		9.7	0.103092784
	1	22.29	0.044863167		10.32	0.096899225
	1	22.04	0.045372051		9.58	0.104384134
#5 READING	1	23.23	0.043047783		9.56	0.10460251
Suspended =150	1	24.69	0.040502228		10.51	0.095147479
40px	1	23.82	0.041981528		9.67	0.103412616
	1	22.08	0.045289855		9.55	0.104712042
	1	22.86	0.043744532		9.5	0.105263158
	1	22.43	0.044583148		9.87	0.101317123
	mean =	22.789	0.043937997		9.883	0.101343845

Figure 7: Raw fall and rise timing measurements for Negative Droplet 5.

D.2.2 Reduced Quantities and Charge Calculation

Droplet #	avg t_d (s)	std dev of t_d (s)	σ of t_d (s)	V_d (m/s)	σ_Vd (s)
1	15.499	1.718666667	0.551707813	6.45203E-05	2.29669E-06
2	15.844	6.518666667	2.063565243	6.31154E-05	8.22032E-06
3	12.554	4.696	1.488032795	7.96559E-05	9.44166E-06
4	17.101	12.79533333	4.047351666	5.84761E-05	1.38397E-05
5	22.789	2.296	0.732230565	4.38808E-05	1.40993E-06

Figure 8: Computed mean fall times, standard deviations, standard errors, and derived terminal fall velocities for negative droplets.

avg t_u (s)	std dev of t_u (s)	σ of t_u (s)	V_u (m/s)	σ_Vu (s)	a (m)	σ_a (m)
3.735	2.43	0.774267396	0.000267738	5.55021E-05	7.79907E-07	1.38809E-08
3.992	1.329333333	0.430943977	0.000250501	2.70421E-05	7.71369E-07	5.02326E-08
4.147	7.834666667	2.479354791	0.000241138	0.000144169	8.6657E-07	5.13575E-08
4.251	3.371333333	1.070321842	0.000235239	5.92287E-05	7.42478E-07	8.78624E-08
9.883	1.167333333	0.381138703	0.000101184	3.90216E-06	6.43179E-07	1.0333E-08

Figure 9: Computed rise velocities, droplet radii, and propagated uncertainties for negative droplets.

Suspended V	E	q (C)	σ_q (C)	corrected a (m)	corrected q (C)	n	e_i
500	83333.3333	8.61792E-19	5.46329E-20	6.87073E-06	1.88445E-14	117618	1.60218E-19
500	83333.3333	8.16648E-19	1.6771E-19	6.86219E-06	1.81543E-14	113310	1.60218E-19
500	83333.3333	8.88567E-19	1.66869E-19	6.95739E-06	1.6593E-14	103565	1.60218E-19
500	83333.3333	7.56765E-19	2.76227E-19	6.8333E-06	1.78132E-14	111181	1.60218E-19
500	83333.3333	2.70019E-19	1.57141E-20	6.734E-06	7.88128E-15	49191	1.60218E-19

Figure 10: Calculated charges q, slip-corrected radii a_{corr}, corrected charges q_{corr}, integer values n, and extracted elementary charge values e_i for negative droplets.

D.3 Positive Polarity Data

D.3.1 Raw Timing Measurements

		FALLING			UPWARDS AT 500V	
	Distance	Time	Velocity(mm/s)		TIME	Velocity(mm/s)
	1	14.71	0.067980965		18.86	0.053022269
	1	17.13	0.058377116		19.57	0.05109862
	1	15.86	0.063051702		19.5	0.051282051
#1 READING	1	16.69	0.059916117		18.99	0.052659294
Suspended =200V	1	16.41	0.060938452		19.6	0.051020408
	1	15.23	0.065659882		19.19	0.052110474
	1	16.46	0.060753341		20.07	0.04982561
	1	15.89	0.062932662		19.97	0.050075113
	1	16.61	0.060204696		19.56	0.051124744
	1	17.28	0.05787037		19.41	0.051519835
	mean =	16.227	0.061768531	mean =	19.472	0.051373842

Figure 11: Raw fall and rise timing measurements for Positive Droplet 1.

		FALLING			UPWARDS AT 500V	
	Distance	Time	Velocity(mm/s)		TIME	Velocity(mm/s)
	1	24.18	0.041356493		37.89	0.026392188
	1	23.94	0.041771094		37.19	0.026888949
	1	25.49	0.039231071		35.01	0.028563268
	1	24.56	0.040716612		36.19	0.027631943
#2 READING	1	28.37	0.035248502		33.08	0.030229746
Suspended =200V	1	25.15	0.039761431		32.04	0.031210986
	1	26.29	0.038037277		34.2	0.029239766
	1	26.41	0.037864445		34.16	0.029274005
	1	26.83	0.037271711		31.29	0.031959092
	1	24.16	0.041390728		38.73	0.025819778
	mean =	25.538	0.039264937	mean =	34.978	0.028720972

Figure 12: Raw fall and rise timing measurements for Positive Droplet 2.

		FALLING			UPWARDS AT 500V	
	Distance	Time	Velocity(mm/s)		TIME	Velocity(mm/s)
	1	17.48	0.057208238		12.85	0.077821012
	1	19.45	0.051413882		12.38	0.080775444
	1	17.48	0.057208238		12.31	0.081234768
#3 READING	1	18.48	0.054112554		15.96	0.062656642
Suspended =200V	1	32.81	0.030478513		15.78	0.063371356
	1	31.93	0.031318509		15.41	0.064892927
	1	33.11	0.030202356		16.05	0.062305296
	1	33.55	0.029806259		17.91	0.055834729
	1	33.81	0.029577048		17.36	0.057603687
	1	35.81	0.027925161		17.23	0.058038305
	mean =	27.391	0.039925076	mean =	15.324	0.066453417

Figure 13: Raw fall and rise timing measurements for Positive Droplet 3.

			FALLING			UPWARDS AT 500V	
		Distance	Time	Velocity(mm/s)		TIME	Velocity(mm/s)
		1	30.49	0.032797639		17.81	0.056148231
		1	32.1	0.031152648		17.56	0.056947608
		1	29.85	0.033500838		16.61	0.060204696
#4 READING		1	34.31	0.029146022		18.25	0.054794521
Suspended =200V		1	28.03	0.035676061		17.16	0.058275058
		1	29.77	0.033590863		16.93	0.059066745
		1	30.82	0.032446463		16.57	0.06035003
		1	31.19	0.032061558		17.19	0.058173357
		1	31.84	0.031407035		16.25	0.061538462
		1	30.41	0.03288392		33.75	0.02962963
		mean =	30.881	0.032466305	mean =	18.808	0.055512834

Figure 14: Raw fall and rise timing measurements for Positive Droplet 4.

			FALLING			UPWARDS AT 500V	
		Distance	Time	Velocity(mm/s)		TIME	Velocity(mm/s)
		1	30.56	0.032722513		17.15	0.058309038
		1	31.24	0.032010243		16.67	0.059988002
		1	34.19	0.029248318		17.28	0.05787037
		1	33.06	0.030248034		18.74	0.053361793
#5 READING		1	32.92	0.030376671		21.31	0.046926326
Suspended =200V		1	16.49	0.060642814		8.7	0.114942529
		1	16.27	0.061462815		8.57	0.116686114
		1	16.79	0.059559261		8.44	0.118483412
		1	16.67	0.059988002		8.47	0.118063754
		1	16.49	0.060642814		7.61	0.131406045
		mean =	24.468	0.045690149	mean =	13.294	0.087603738

Figure 15: Raw fall and rise timing measurements for Positive Droplet 5.

D.3.2 Reduced Quantities and Charge Calculation

Droplet #	avg t_d (s)	std dev of t_d (s)	σ of t_d (s)	V_d (m/s)	σ_Vd (s)
1	16.227	2.145333333	0.685014971	6.16257E-05	2.6015E-06
2	25.538	3.832	1.215492657	3.91573E-05	1.86371E-06
3	27.391	24.44933333	7.732140069	3.65083E-05	1.03059E-05
4	30.881	3.944	1.250805181	3.23824E-05	1.31162E-06
5	24.468	26.42	8.355276177	4.08697E-05	1.39561E-05

Figure 16: Computed fall velocities and associated uncertainties for positive droplets.

avg t_u (s)	std dev of t_u (s)	σ of t_u (s)	V_u (m/s)	σ_Vu (s)	a (m)	σ_a (m)
19.472	0.958666667	0.31765418	5.13558E-05	8.37787E-07	7.62211E-07	1.60882E-08
34.978	6.746666667	2.135591513	2.85894E-05	1.74553E-06	6.07577E-07	1.44589E-08
15.324	5.621333333	1.780151354	6.52571E-05	7.58076E-06	5.86666E-07	8.28042E-08
18.808	9.961333333	3.151478411	5.31689E-05	8.909E-06	5.52521E-07	1.11897E-08
13.294	16.45333333	5.203865657	7.52219E-05	2.94452E-05	6.20719E-07	1.05981E-07

Figure 17: Computed rise velocities, droplet radii, and propagated uncertainties for positive droplets.

Suspended V	E	q (C)	σ_q (C)	corrected a (m)	corrected q (C)	n	e_i
500	83333.3333	1.68293E-19	1.23395E-20	6.85303E-06	3.80874E-15	23772	1.6022E-19
500	83333.3333	7.57135E-20	6.16257E-21	6.6984E-06	2.40677E-15	15022	1.60216E-19
500	83333.3333	1.81401E-19	7.8625E-20	6.67748E-06	6.07708E-15	37930	1.60218E-19
500	83333.3333	1.23177E-19	8.71554E-21	6.64334E-06	4.51452E-15	28177	1.6022E-19
500	83333.3333	2.26513E-19	1.18289E-19	6.71154E-06	6.97313E-15	43523	1.60217E-19

Figure 18: Calculated charges q, slip-corrected radii, corrected charges q_{corr}, integer multiples n, and extracted elementary charge values e_i for positive droplets.